

Bitcoin Market Sentiment Analysis Using Hyper liquid Trading Data

Submitted by

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Objective

The objective of this project is to analyze the relationship between Bitcoin market sentiment (Fear & Greed Index) and trader performance using Hyperliquid historical trading data. The analysis focuses on identifying patterns in profitability under different market sentiments and building a machine learning model to predict profitable trades.

Dataset Description

Dataset 1

Historical Trader Data

Contains over 211,224 cryptocurrency trades.

Columns include:

- Account
- Coin
- Execution Price
- Size USD
- Side
- Closed PnL
- Fee
- Timestamp

Dataset 2

Fear & Greed Index

Contains:

- Date
- Fear & Greed Score
- Market Classification

Data Preprocessing

Data Cleaning

- Loaded both datasets using Pandas.
- Converted UNIX timestamps into readable datetime format.
- Created a common date column.
- Merged both datasets using the date column.
- Checked missing values.
- Removed invalid records.
- Created a Profit column for machine learning

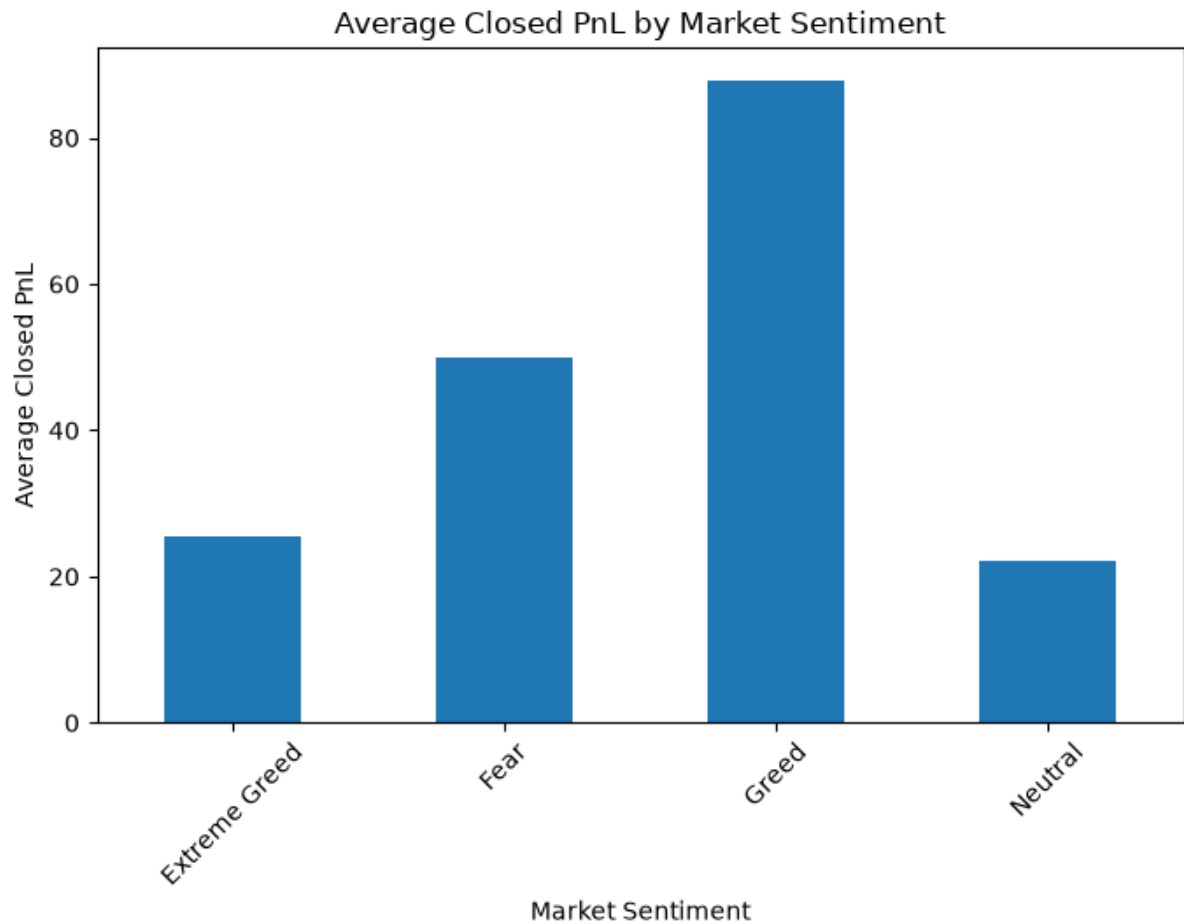
Exploratory Data Analysis (EDA)

The following analyses were performed:

- Missing Value Analysis
- Sentiment Distribution
- Average Closed PnL
- Average Trade Size
- Profit Distribution
- Correlation Analysis

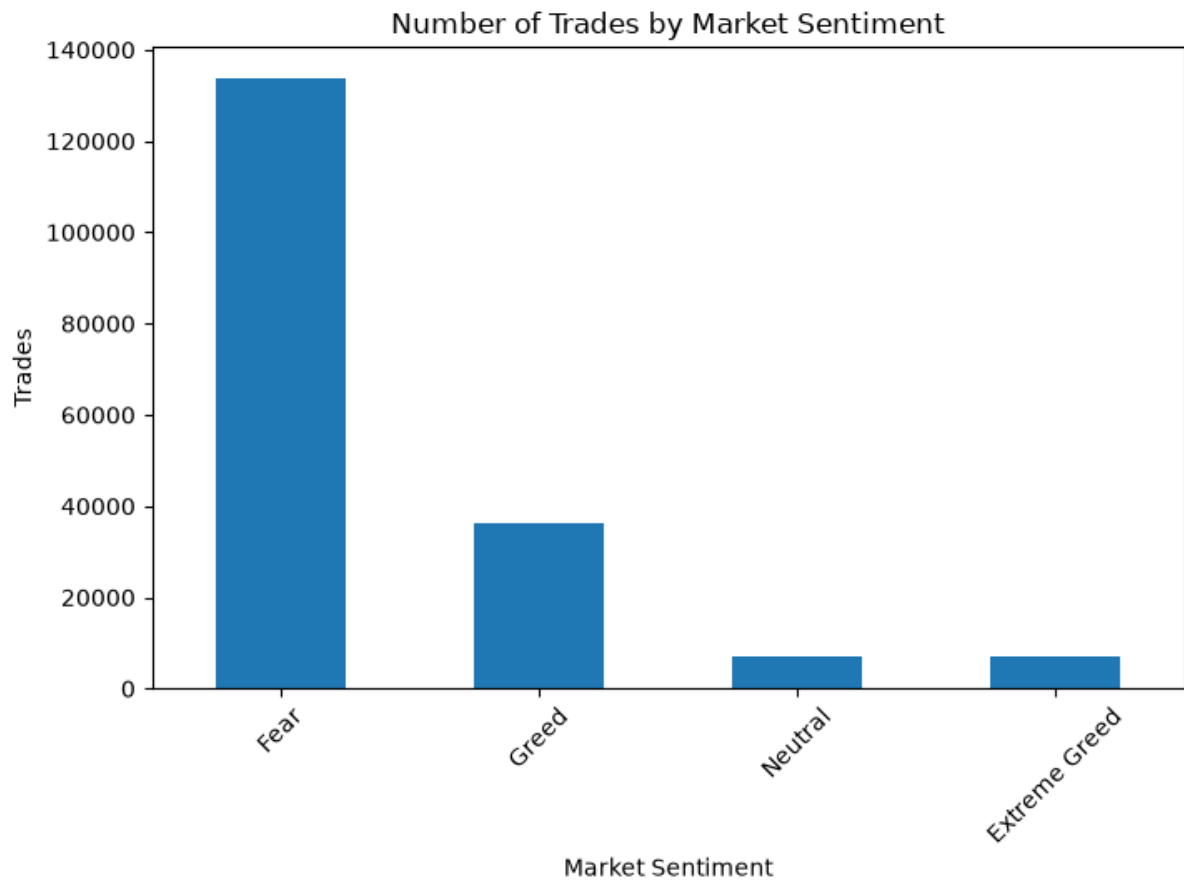
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Average Profit by Market Sentiment



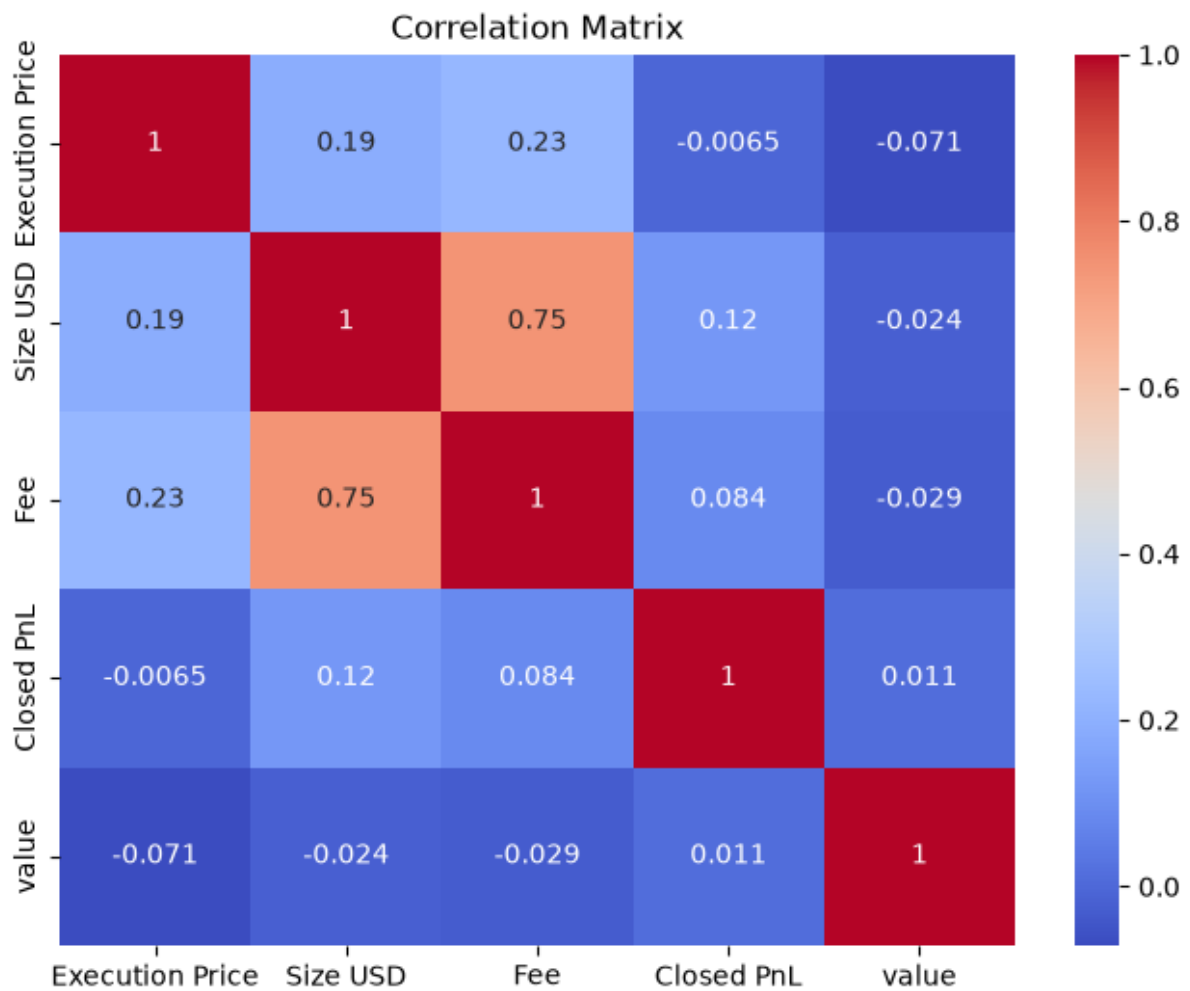
Traders earned the highest average profit during **Greed** market conditions, followed by **Fear**. Neutral market conditions produced the lowest average profitability.

Number of Trades by Market Sentiment



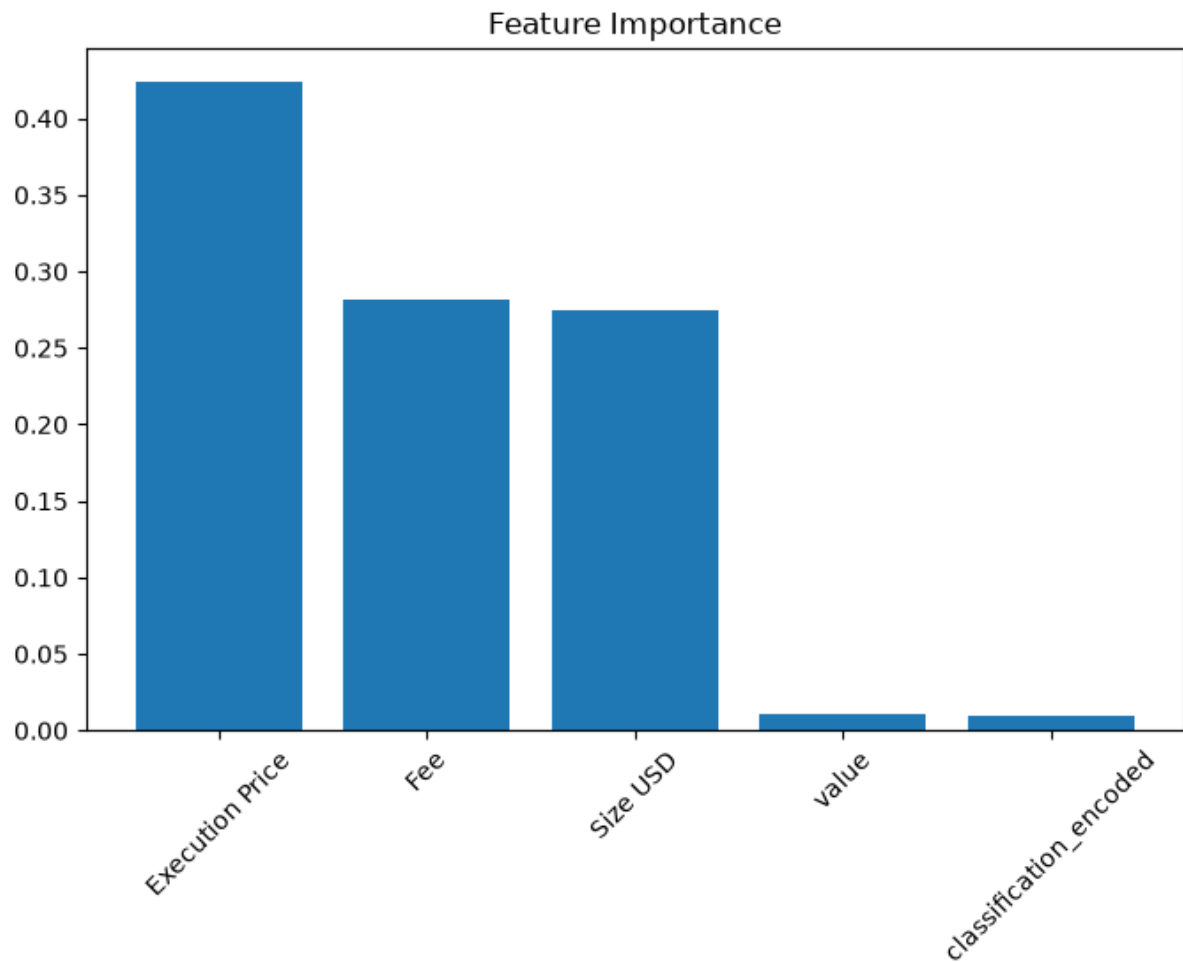
Most trades occurred during Fear market conditions, indicating higher trading activity during uncertain market periods.

Correlation Heatmap



The heatmap illustrates the relationships among numerical variables. Execution Price, Trade Size, and Fees show stronger relationships with profitability than the Fear & Greed Index value.

Feature Importance



Execution Price was identified as the most important feature influencing trade profitability, followed by Trading Fee and Trade Size.

Machine Learning Model

Random Forest Classifier

Features Used

- Execution Price
- Size USD
- Fee
- Fear & Greed Value
- Market Classification

Target

Profit (0/1)

Results

Metric	Value
Accuracy	81.76%
Precision	0.82
Recall	0.81
F1 Score	0.81

Confusion Matrix

[[18419 2919]

[3804 11711]]

The Random Forest model correctly classified the majority of profitable and non-profitable trades, achieving an overall accuracy of 81.76%.

Key Findings

- Analyzed more than **211,000** cryptocurrency trades.
- Successfully merged trading data with the Fear & Greed Index.
- Greed market conditions showed the highest average profitability.
- Execution Price was the strongest predictor of profitable trades.
- Trading Fee and Trade Size significantly influenced profitability.
- Fear & Greed Index had a measurable but comparatively smaller influence.

Conclusion

This study demonstrates that Bitcoin market sentiment provides useful context for understanding trader performance. However, trade-specific characteristics such as execution price, trading fees, and trade size have a stronger impact on profitability. The Random Forest model achieved an accuracy of 81.76%, indicating that combining market sentiment with trading metrics can effectively predict profitable trades

